

Campus Wayfinding Study

Overview

Students and visitors navigate using signage, mobile maps, and habit.

Prior work suggests people scan for a single confirming detail.

This study asks where that scanning breaks down, and whether the failures cluster in places that can be predicted in advance.

Scope and Definitions

A junction is any point offering two or more onward paths.

A backtrack is a return to a previously visited junction.

First correct turn is the first heading change that reduces remaining distance to the stated destination.

Outdoor routing and emergency egress are out of scope.

Methods

Participants and procedure

Twenty-four participants, none familiar with the study building.

Four timed trials each, destinations presented in random order.

Participants thought aloud while a researcher logged junction events.

Audio was transcribed and coded for expressions of uncertainty.

Findings

Where navigation failed

Median time to first correct turn was eleven seconds.

The distribution was strongly right skewed at the entrance junction.

Backtracking concentrated at three junctions, all unsigned, all offering three or more onward paths.

Two-path junctions produced almost no backtracking.

The Single Sign Intervention

One directional sign at the worst junction halved backtracks there.

The effect did not transfer to other junctions.

This supports a decision-point account over signage quantity.

Limitations

One building, twenty-four participants, one signage style.

Think-aloud protocols may themselves slow navigation.

We did not test whether the effect survives familiarity.